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Paleomagnetism of the Khorat Group, Mesozoic, Northeast Thailand

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Abstract—The Khorat Group, middle or late Triassic to Cretaceous, consists of up to 5000 m of paralic and freshwater sedimentary deposits, mainly alluvial flood-plain red-beds. Paleomagnetic measurements tested using thermal and AF demagnetization by several authors in different laboratories are generally consistent. Selection of samples requires care because of deep weathering in the monsoon climate of Thailand. All authors have reported a clockwise rotation of the paleomagnetic declinations, which we estimate to be 37 ± 7 degrees, and which occurred at some time in the last 100 My. Inclinations are not significantly different from those of the present day, indicating that Northeast Thailand (and the Indochina plate) have remained at nearly the same latitude since about late Triassic. Relative southward movement of the South China plate possibly resulted from dextral displacement along the Red River Fault. Relative southward movement of Eurasia involved either ocean consumption or continental crust shortening of 1000 km or more. Magnetostratigraphy suggests an unconformity, supported by stratigraphic evidence, between Sao Khua and Phu Phan formations in the upper part of the Khorat Group, and the unconformity is explained by a simple model involving the combined effect of subsidence at an exponentially decreasing rate, and Mesozoic eustatic sea-level changes.

INTRODUCTION

THE FOLLOWING is chiefly a discussion of paleomagnetic data reported by Somkiat Maranate (1982a,b). Other paleomagnetic data from Northeast Thailand (Haile and Tarling 1975, Barr *et al.* 1978, Bunopas *et al.* 1978, Bunopas 1981, and Achache and Courtillot 1985) are mentioned also. There are now a substantial number of paleomagnetic measurements from the region, made in various laboratories, and generally in good agreement, lending some confidence to tectonic inferences based on them.

SOURCES AND RELIABILITY OF PALEOMAGNETIC DATA

Maranate (1982a,b) measured directions and intensities of magnetisation in three samples from each of 179 surface sampling sites in Khorat Group sediments and five specimens taken at 5 m intervals from drill cores from Nakhon Phanom (Fig. 1). The Khorat Group has an estimated maximum thickness of approximately 5000 m, ranges from late Triassic to middle Cretaceous, consists mainly of alluvial flood-plain red-beds, and outcrops over most of the Khorat Plateau of Northeast Thailand. The surface samples were taken from six widely separated sections (Fig. 1) and represent all formations of the Khorat Group except the basal Huai Hin Lat Formation (Norian) and its equivalents. Some of the samples were from Lom Sak Tuff, of uncertain age, thought by us to be probably middle Triassic and doubtfully part of the Khorat Group.

There is little or no paleontological correlation between the sections, identified fossils being restricted to one or a few localities in any formation. Lithological correlation is excellent; strong colour and grain-size contrasts make it possible to trace formations for hundreds of kilometers and structures are simple with gentle to moderate dips at most places. Sites representing the same formation in different sections cannot usually be placed in relative stratigraphic order, but can be assigned to approximate stratigraphic horizons with reasonable confidence. Precise section to section correlation would be desirable but is not critical for the middle Triassic to middle Jurassic and for a large part of the Cretaceous because few reversals occurred and consequently they can be identified with approximate independent dating control.

All the paleomagnetic measurements are recorded in Maranate (1982a) and the site means and statistical data are repeated in Maranate (1982b). One specimen from each site was tested by stepwise thermal demagnetization, up to 600°C in some cases; many specimens developed high intensity viscous magnetization after being heated to 550 or 500°C and a few even after heating to 450°C. It was inferred that chemical change caused by the heating had produced a new magnetic mineral phase. Diagrams (Fig. 2) show the highest

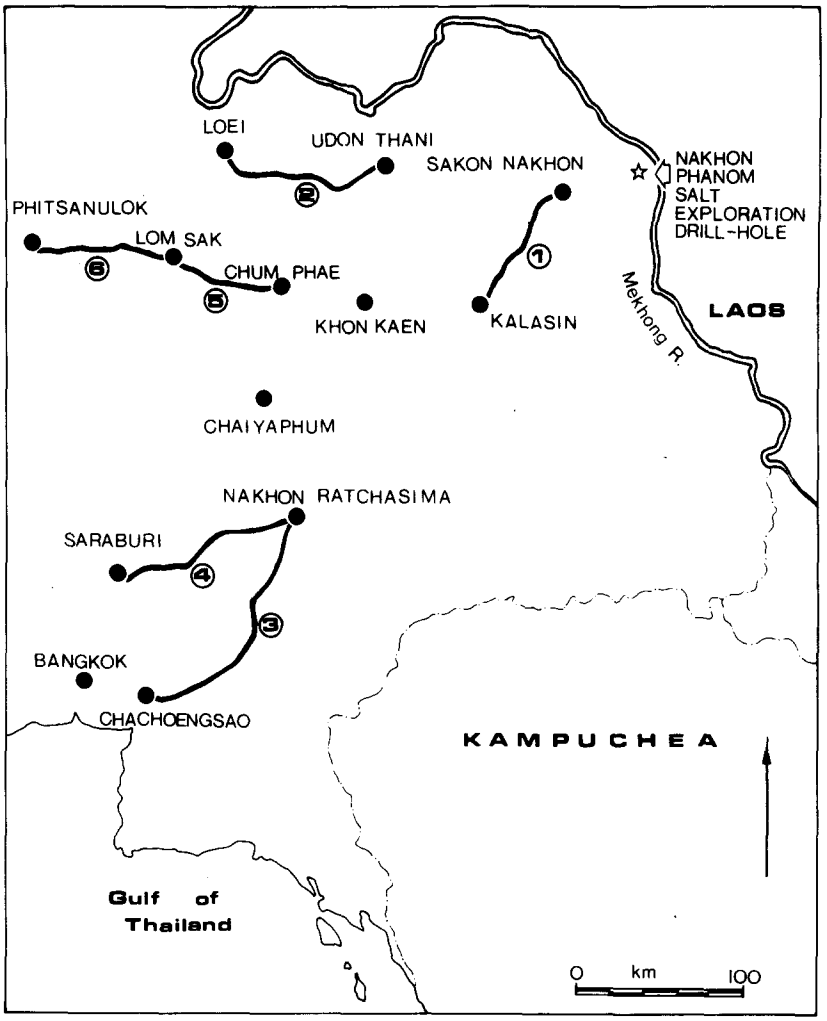


Fig. 1. Map of Northeast Thailand showing sections from which Maranate (1982a,b) collected paleomagnetic samples.

temperature reached before the appearance of the new unstable magnetization. The appropriate cleaning temperature for the other two specimens of each site was chosen according to the behaviour of the pilot sample. For paleomagnetic stratigraphy 36 sites were rejected because of erratic direction and/or intensity changes during demagnetization of the pilot sample or because in-site agreement of the three specimens was too poor to be certain of the polarity. For determining paleopole positions 105 sites were discarded; only those with in-site alpha-95 of 20 degrees or less were accepted. Thailand is subject to lateritic weathering and we consider that many samples have hard components of magnetization formed by weathering during the Quaternary and perhaps Cenozoic. The Nakhon Phanom drill cores all gave excellent demagnetization results and we presume that is because they have not been subjected to weathering.

Orthogonal vector projections illustrating the demagnetization behaviour of selected specimens (Fig. 2A–L) show differences from site to site, probably according to differing dominant magnetic minerals. Petrographic examinations by Maranate (1982a) showed that the dominant mineral is sometimes very finely divided haematite, sometimes detrital titanomagnetite and perhaps sometimes detrital magnetite. Abundant sources of detrital titano-magnetite and magnetite are the extensive andesites, metabasites, rhyolites and granites in the north–south trending Pha Som orogenic belt in central Thailand (Bunopas 1981) immediately west of the Khorat Plateau.

Haile and Tarling (1975) and Barr *et al.* (1978) used thermal cleaning on their samples. Bunopas (1981) used AF cleaning and did not have a strong enough field to randomize the hard magnetization of haematite. Thus some of his measured directions were not well verified. Nevertheless his mean directions except for the Cretaceous one are in good agreement with those of other authors (Table 2). Achache and Courtillot (1985, Fig. 3) used both thermal and AF demagnetization to test late Triassic samples from the Huai Hin Lat Formation ('grey beds' at the base of the Khorat Group). The eight specimens they illustrate all show large soft components of magnetization like the titanomagnetite dominated samples of Maranate (1982a), but after cleaning at 200–300°C or 20–30 mteslas they show weaker stable components with consistent directions. Achache and Courtillot thus have determined a reliable upper Triassic paleopole position for Thailand (Table 2, h) for the first time.

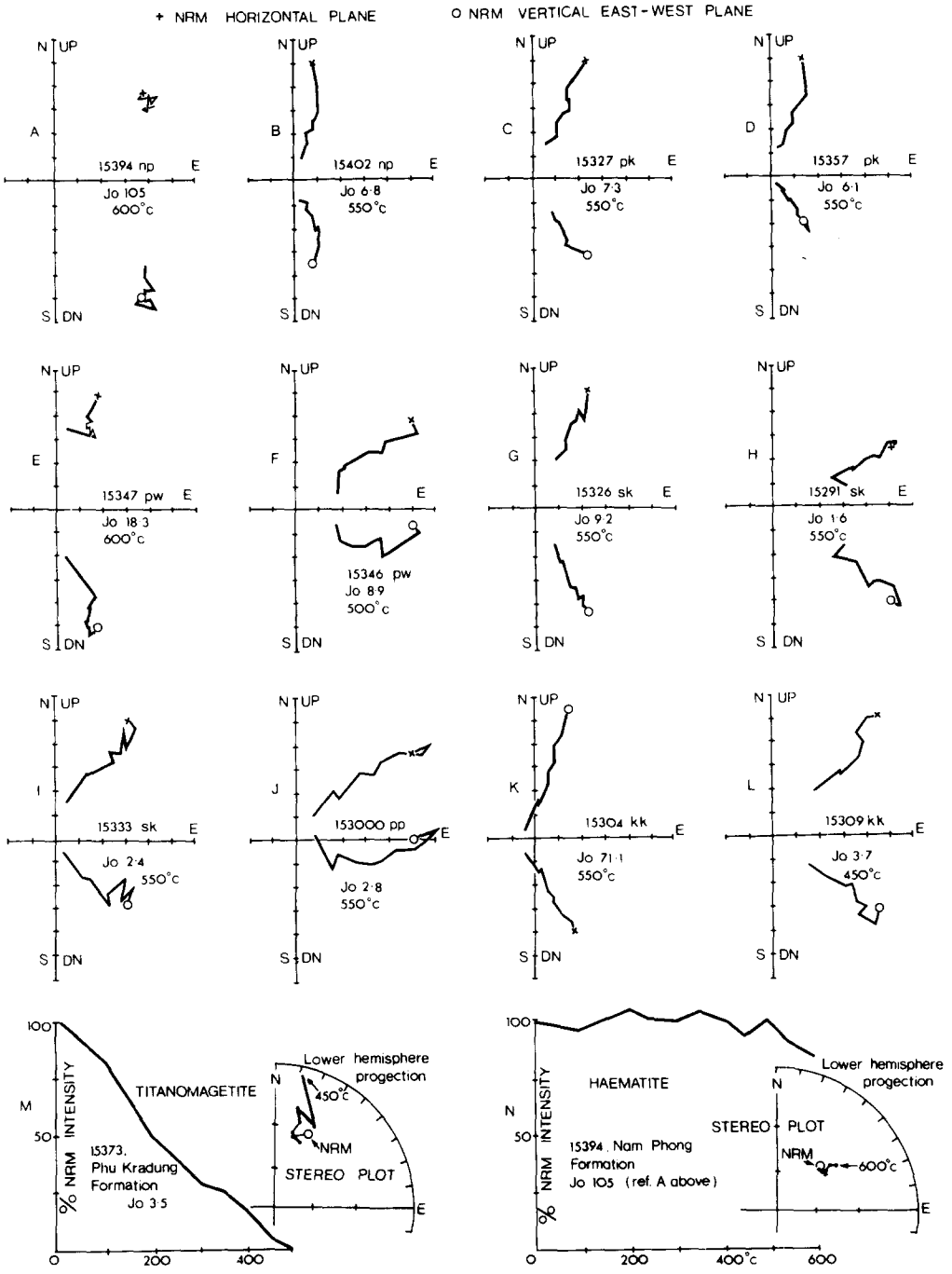


Fig. 2. Selected demagnetization diagrams for Khorat Group specimens. A to L orthogonal vector projections for 12 pilot specimens; each diagram shows Jo (NRM intensity) and final cleaning temperature (if less than 600°C, the last before development of strong viscous component—see text); note consistent declination directions in northeast quadrant except for 15304 (third row center right) which has reversed polarity and tectonically corrected declination of 166° after cleaning at 350°C. M and N, contrast between magnetization of detrital titanomagnetite and single-domain haematite dominated rocks.

PALEOMAGNETIC STRATIGRAPHY

Stratigraphic control on sampling positions is not good enough to integrate the polarity determinations of all authors. Therefore our magnetostratigraphy (Fig. 3, right) is limited to the results of Maranate (1982a,b) and Achache and Courtillot (1985). The latter authors defined one reversed and four normal horizons in Huai Hin Lat Formation and quoted recently published paleontological evidence for a Norian age. The one reversed horizon is itself based on an important Norian vertebrate locality (Buffetant *et al.* 1981). We have shown the Huai Hin Lat Formation as predominantly normal and have arbitrarily placed the reversed horizon at about the middle (Fig. 3).

On the left side of Fig. 3 is a magnetostratigraphic time-scale, the upper part fairly reliable, being based on ocean-floor magnetic anomalies (Larson and Hilde 1975, Van Hinte 1976), the lower part highly unreliable, being based on descriptions of land sequences, mainly red-beds (Creer 1971, McElhinny and Burek 1971, Pechersky and Kramov 1973, Irving and Pullaiah 1976). Quantitative

Table 1. Paleomagnetic pole positions reported by Maranate (1982a)

	Fm	Age	N	D	I	k	A_{95}	°N	°E	dP	dM
1	kk	K	7(21)	047	29	19	14	45	189	9	16
2	pp	Kl?	3(9)	045	19	10	41	46	197	23	43
3	sk	Jm	16(48)	033	35	23	8	59	184	5	9
4	pw	Jm	14(42)	031	41	10	13	60	174	10	16
5	pk	Jl	15(45)	033	29	31	7	58	188	4	8
6	np	Tu-Jl	9(27)	034	41	20	12	58	174	9	14
7	ls	Tm?	10(30)	018	21	13	14	71	208	8	15
8	Mean for Khorat Group excluding Lom Sak Formation (uppermost Triassic-mid Cretaceous):										
				037	32	63	9	54	186	5	10
9	Mean including Lom Sak Formation (mid Triassic?-mid Cretaceous):										
				034	31	45	9	57	188	6	10

Fm Formation: kk Khok Kruat, pp Phu Phan, sk Sao Khua, pw Phra Wihan, pk Phu Kradung, np Nam Phong, ls Lom Sak.

N number of sites, number of specimens in brackets. D, I mean declination and inclination. °N, °E paleopole position. dP, dM error limits of paleopole position.

Note: In Khorat Plateau present declination is 359.75 degrees, and inclination ranges from 11 degrees in the south to 18 degrees in the north.

Table 2. Thailand Mesozoic pole positions reported by other authors

a-c, Barr et al. (1978). d-g, Bunopas (1982). h, Achache and Courtillot (in press).

	Age	N	D	I	k	A_{95}	°N	°E	dP	dM
a	Jm-u	9(?)	002	23	13	26	80	111	43	81
b	Tm-Jl	10(?)	036	39	32	12	56	176	9	14
c	Tm-Ju	19(?)	026	36	15	15	64	171	10	17
d	K	2(6)	010	34	36	43	79	176	28	49
e	Jm-u	14(36)	029	46	10	14	67	177	11	17
f	Jl	16(44)	039	36	19	9	53	180	6	10
g	Tu-Jl	8(24)	030	20	11	18	60	185	10	19
h	Tu	39(?)	043	43	106	8	49	172	A_{95}=7	

See Table 1 for explanation of symbols. Note: According to our interpretation there is no Upper Jurassic in the Khorat Group (see section on paleomagnetic stratigraphy below).

ages have been adjusted to the IGCP Project 133 time-scale (Odin 1982). The reason for showing the pre-sea-floor spreading part of the time-scale is to show the general similarity to the Khorat Group sequence of polarity changes. In constructing Fig. 3 we made no attempt to place Triassic to Middle Jurassic reversed intervals in the Khorat Group sequence on the right adjacent to those in the time-scale on the left, except for the single reversed site at the base of Nam Phong Formation, which, on stratigraphic grounds, we consider very likely to represent the inferred reversed or mixed polarity interval in the Rhaetian.

There are three strongly contrasted magneto-stratigraphic horizons in the middle Triassic to Cretaceous, the top and bottom of the middle Cretaceous quiet normal interval dated at about 80 My and 115 My (Santonian and upper Hauterivian) and the bottom of the late Jurassic–early Cretaceous mixed polarity interval dated at about 150 My (uppermost Callovian). Besides them there are a number of paleontologically dated horizons in the Khorat Group (Fig. 3), three of which give fairly definitive horizons in the Sao Khua (upper Callovian or lower Oxfordian), Phu Kradung (lower Jurassic) and Huai Hin Lat Formation (Norian).

Starting from the top of the sequence (Fig. 3) no satisfactory polarity determinations have been obtained from the Maha Sarakham ('Salt') Formation, which could be upper Cretaceous or Tertiary. Esso Petroleum reported seismic evidence of an unconformity between the Maha Sarakham and underlying Khok Kruat Formation at a seminar in the Department of Mineral Resources, Bangkok, on 2 February 1983.

Seven species of brackish or freshwater bivalves from the Khok Kruat Formation indicate a middle Cretaceous age (Iwai *et al.* 1966). The four reversed sites in the base of Khok Kruat and top of Phu Phan Formation (Fig. 3) indicate a reversed polarity interval that is unusually long for early-mid Cretaceous and probably represents the 116–119 My interval in the Hauterivian. Consequently the ten consistently normal sites higher in the Khok Kruat Formation are taken as representing the middle Cretaceous quite normal interval, though that still needs to be verified by closing the rather large sampling gap (Fig. 3). The Phu Phan Formation is probably conformable with the Khok Kruat Formation and the few polarity determinations from it are consistent with the rapidly alternating polarities of the upper Jurassic and lower Cretaceous. The Phu Phan Formation, however, is thin (maximum 190 m) and is unlikely to represent the entire 30 My from Oxfordian to Valanginian. It consists of medium to coarse sandstones and conglomerates which contain some red-bed clasts and we infer unconformity between it and the underlying Sao Khua Formation which consists of fine sandstones and shales, mostly purple-red in colour.

Twenty four sites from the top of Sao Khua to the top of Phra Wihan Formation (23 from Sao Khua) are all normal. The Sao Khua samples gave good demagnetization results, directions of magnetization with good in-site and in-formation agreement (K23, A95 8 for formation), different from the present field direction, and acceptable with confidence (Maranate 1982a,b). The sites are spread fairly evenly through the formation with an average stratigraphic interval of about 30 m, allowing for the possibility of undiscovered brief reversed intervals, but not for the rapid polarity alternations characteristics of the late Jurassic and early Cretaceous. The age of the top of the formation therefore must be greater than 150 My and, with the fossil evidence, the upper part of the formation must be Callovian (Fig. 3). Provisionally we show the Phu Phan-Sao Khua unconformity as representing the whole of the upper Jurassic and possibly basal Cretaceous.

Below we discuss a possible regime of tectonic subsidence and eustatic sea-level change that would account for the unconformity.

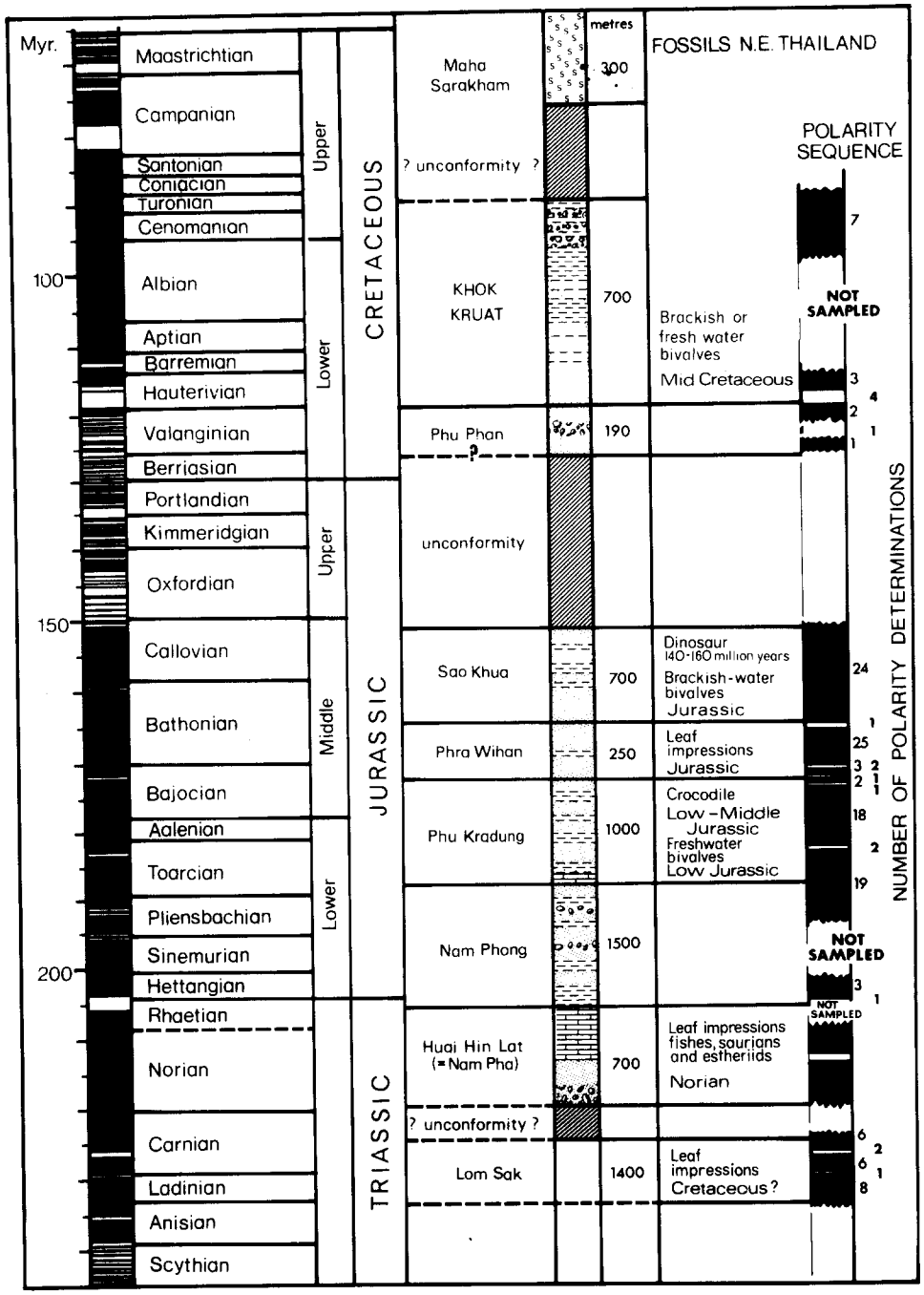


Fig. 3. Stratigraphic sequence of Khorat Group formations showing generalized lithologies and paleontology, maximum estimated thicknesses, and geological correlations (see text). On left is ocean-floor spreading magnetostratigraphy down to Callovian and poorly defined land-based magnetostratigraphy for the rest of the Jurassic and the Triassic; on right the magnetostratigraphy proposed here for Northeast Thailand. On extreme right, figures in left column show numbers of normal polarity sites, while those in right column show numbers of reversed polarity sites.

The lower formations of the sequence, Phra Wihan, Phu Kradung, Nam Phong and Huai Hin Lat all have predominantly normal polarities with one or a few short reversed intervals in each, as expected for upper Triassic to middle Jurassic. The potential value of the widely spaced reversed intervals for precise correlation of individual horizons is evident because widely spaced polarity reversals are easier to identify than the closely spaced ones in mixed polarity intervals. Their value will be greatly enhanced by better paleonotological control, which we can look forward to because substantial advances are being made in Khorat Group paleontology, particularly by Mrs Rucha Ingavat, Dr E. Buffetaut and their colleagues.

Iwai *et al.* (1966) considered Lom Sak Tuff to be Cretaceous on the evidence of poorly preserved plant fossils identified as *Sequoia ambigua*, but Bunopas (1971) mapped it as underlying Nam Phong Formation at the Phitsanulok–Lom Sak Highway, and Triassic in age. It is considerably more deformed than the Nam Phong Formation and presumably unconformable. It has a basic to intermediate composition not consistent with extensive late Triassic and early Jurassic acidic vulcanism in Thailand, and predominantly normal polarities.

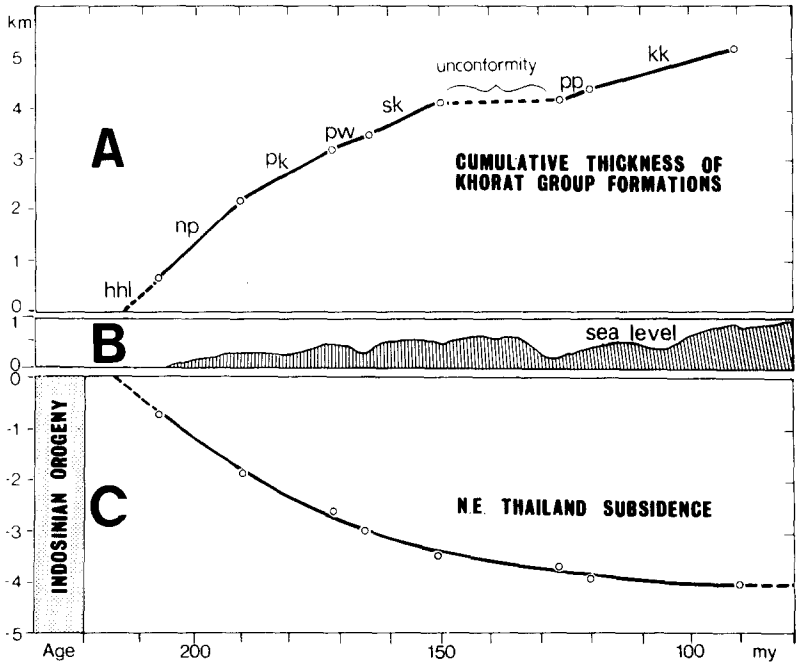


Fig. 4. Subsidence of Northeast Thailand at exponentially decreasing rate during Jurassic and Cretaceous. A: Cumulative thicknesses at inferred ages of formation tops (Fig. 3; Lom Sak Tuff not included, see text). B: Jurassic–Cretaceous sea-level changes (see text for sources). C: Jurassic–Cretaceous subsidence of Northeast Thailand derived by subtracting B and A.

We have arbitrarily placed it below Huai Hin Lat Formation at about Ladinian or Carnian.

INFERRED TECTONIC EVENTS

1. *Subsidence of Northeast Thailand*

The Khorat Group is a post-orogenic molasse facies (Bunopas 1981, and see next section below). Most of it was deposited on a large alluvial plain, probably similar to the present Indogangetic Plain, and some parts were deposited in paralic conditions (Bunopas 1981). The estimated maximum thickness is approximately 5000 m and indicates progressive subsidence of Northeast Thailand between late Triassic and Cretaceous. If vertical crustal movement were the only factor, the Phu Phan-Sao Khua unconformity would indicate that subsidence was interrupted by still-stand or uplift between late Jurassic and early Cretaceous. However, if allowance is made for eustatic sea-level changes during the Jurassic (Hallam 1978) and Cretaceous (Sliter 1976, Hancock and Kaufman 1979) the subsidence appears to have been continuous from late Triassic to Cretaceous and decreasing in rate exponentially (Fig. 4C). Hancock and Kaufman (1979) estimated a net eustatic sea-level rise of 600 m between 105 and 70 My ago (beginning of the Albian to Maastrichtian) and we consider it reasonable to assume a net rise of 1000 m between the beginning of the Jurassic and the Maastrichtian (Fig. 4B). The points that define the subsidence curve (Fig. 4C) were derived by subtracting sea-level (above basal Jurassic sea-level) at the appropriate time from each point defining the cumulative thickness curve (Fig. 4A). Using values of 800 and 1200 m for the Jurassic to Maastrichtian net sea-level rise gives essentially exponential subsidence curves, but using a value of 600 m fails to remove the effect of the Phu Phan–Sao Khua unconformity.

Our data points are too widely spaced in time to show effects of short term sea-level changes on the subsidence curve, which shows significant smoothing of irregularities on the cumulative thickness curve at only two places. The larger one is between 150 and 125 My where the effect of the Phu Phan–Sao Khua unconformity disappears, suggesting that the unconformity was formed as a result of the late Jurassic (Kimmeridgian–Tithonian) fall of sea-level (Hallam 1978). The lesser smoothing effect is the reduction of the dip in the cumulative thickness curve caused by the thinness of the Phra Wihan (pw) Formation relative to the subjacent and superjacent Phu Kradung (pk) and Sao Khua (sk) formations. The Phra Wihan is not only thinner, but also coarser-grained, and is a prominant ridge-former. It probably owes its distinctive lithology to the 170–160 My (Bathonian) low sea-level (Hallam 1978).

Errors in age, thickness, and sea-level estimates are sufficient to account for the scatter of the datum points on Fig. 4C. The subsidence shown is possibly exaggerated because it is based on the maximum measured thickness of formations, and the true maximum thickness of the Khorat Group at any one place might be nearer 4000 m than 5000 m; unpublished petroleum exploration results may soon yield more certain thickness values.

Lom Sak Tuff Formation was not included in the plots (Fig. 4) because it is probably older than the Indosinian Orogeny, is localized in distribution, and in our opinion probably should not be treated as a part of the Khorat Group.

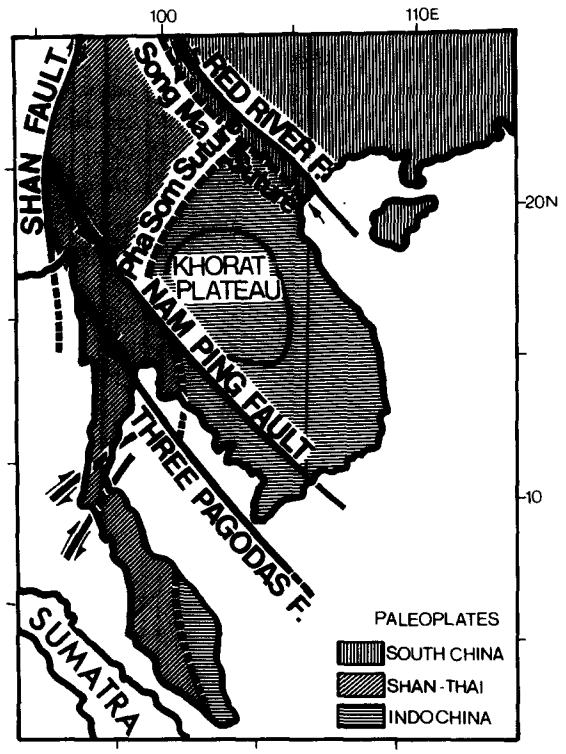


Fig. 5. Map of continental Southeast Asia showing position of the Khorat Plateau (Northeast Thailand) and major tectonic features: the three paleoplates, South China, Shan-Thai and Indochina, the Song Ma and Pha Som (late Triassic) geosutures between them, and post-Triassic strike-slip faults. The Shan and Red River Faults are known to be active and dextral. The Nam Ping and Three Pagodas faults are possibly no longer active and are sinistral.

The exponential decay of the subsidence rate of Northeast Thailand is like that of ocean lithosphere which gradually sinks isostatically because of cooling and thermal contraction (Sclater *et al.* 1971). Average subsidence rates estimated from individual formation thicknesses and age ranges in the Khorat Group decrease from 4 m/My in the early Jurassic to 0.7 m/My in the middle Cretaceous. The fastest is ten times slower than the average subsidence rate of ocean lithosphere less than 60 My old (Sclater and Tapscott 1979), but the eventual amount of subsidence is similar. We postulate that the lithosphere of Northeast Thailand gradually cooled after the mid to late Triassic Indosinian Orogeny (Bunopas 1981) and sank isostatically because of thermal contraction combined with the load of Khorat Group sediments.

2. *Lateral displacement relative to South China*

Continental Southeast Asia is part of a tectonic collage (Mitchell 1981) and contains two main paleoplates, Indochina and Shan-Thai (Fig. 5) that were sutured to each other and to South China in the Indosinian Orogeny (Bunopas and Vella 1978, Gatinsky *et al.* 1978, Bunopas 1981). Paleozoic and lower to middle Triassic continental margin marine sediments and basic to intermediate volcanics between the paleoplates are overlain by very extensive upper Triassic and younger continental deposits mapped in Malay Pen-

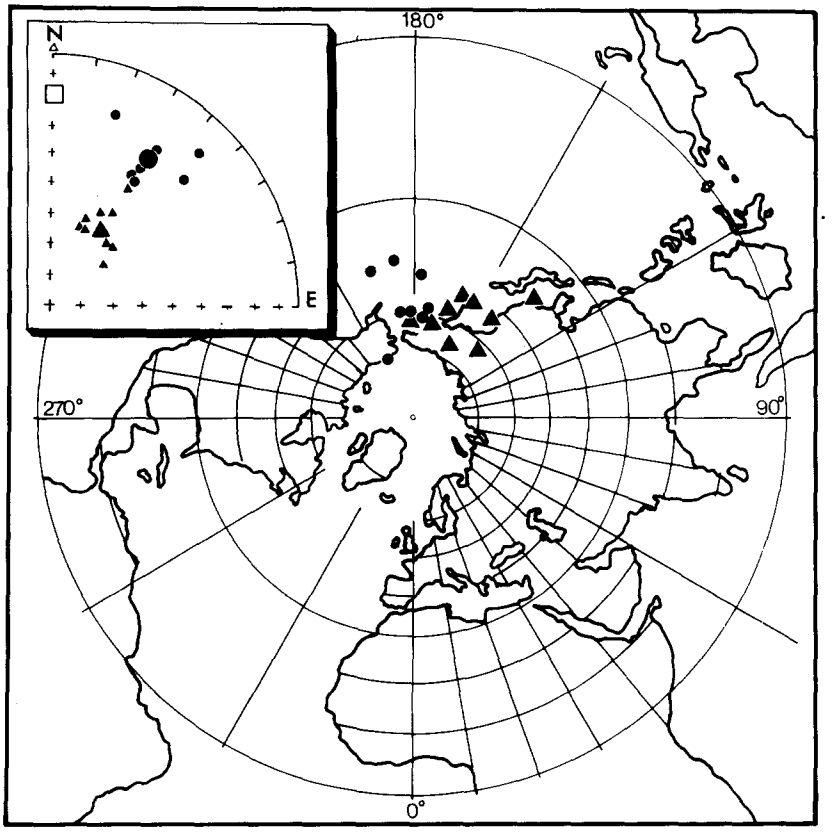


Fig. 6. Map showing Mesozoic pole positions for Northeast Thailand (circles) and South China (triangles). Northeast Thailand is part of the Indochina paleoplate. Inset: Equal area projection of Northeast Thailand (circles) and South China (triangles) mean directions of magnetization from which pole positions were calculated. Opposites of reversed polarities were used when calculating the means. Large circle and large triangle indicate overall mean directions; declinations are the same for both regions but inclinations are significantly different.

Table 3. Comparison of Northeast Thailand and South China paleolatitudes

Fmtn	Age	PDl °N	I	Pl °N	Delta-1	Mean delta-1
NORTHEAST THAILAND:						
kku	Km	17.3	38°	21.3	-4.0°	
kk	Kl-m	17.3	29	15.5	+1.8	
pp	Kl?	17.3	19	9.8	+7.5	
sk	Jm	17.0	35	19.3	-2.3	-1°(alpha-95 5)
pw	Jm	16.5	41	23.5	-6.8	-2°(alpha-95 5)
pk	Jl	16.5	31	16.7	+0.3	with Lom Sak
np	Tu-Jl	17.0	41	23.5	-6.5	Tuff excluded.
hhl	Tu	17.0	43	24.9	-7.9	
ls	Tm?	17.0	21	10.9	+6.1	
Kwantung:						
-	K	23.3	56	36.5	-13.2	
-	K	23.3	55	35.5	-12.2	
-	K	23.3	60	40.9	-17.6	-11°(alpha-95 11)
Yunnan:						
-	K	25.0	45	26.5	-1.5	
Yunnan:						
-	J	25.0	64	45.7	-20.7	
Szechuan:						
-	J	28.5	64	45.7	-17.2	-20°(alpha-95 5)
-	J	29.0	64	45.7	-16.7	
-	J	29.0	69	52.5	-23.5	

hhl Huai Hin Lat, other Thailand formation symbols as on Table 1; kku upper Khok Kruat from Nakhon Phanom drill cores. PDl present day latitude, Pl paleolatitude, delta-1 present latitude minus paleolatitude.

T-test confidence levels of mean delta-1 differences: South China Jurassic-Cretaceous 90%; Northeast Thailand-South China Cretaceous 98%; Northeast Thailand-South China Jurassic 99.9%.

insula, Western Thailand, North Thailand, Northeast Thailand (Khorat Group), Indochina, Yunnan and Szechuan. This well established stratigraphic evidence shows that sea had disappeared from between Shan Thai, Indochina and South China and continent–continent collision was complete by late Triassic (Bunopas 1981).

Clockwise rotation of Khorat Group paleomagnetic declinations has been known for some time (Haile and Tarling 1975, Barr *et al.* 1978, Bunopas 1981). Maranate estimated it to be 37 ± 7 degrees with no stratigraphically significant difference between Khorat Formations down to Nam Phong (lower Jurassic) (Table 1; Fig. 6). In other words that rotation has occurred within the last 100 My. The mean of South China Jurassic and Cretaceous declinations (Lee *et al.* 1963, Chen *et al.* 1965, as listed by McElhinny 1969) is almost the same (Fig. 6). A systematic difference between Northeast Thailand and South China paleopole positions (Maranate 1982b) results from different paleomagnetic inclinations (Fig. 6; Table 3). Inclinations in the Khorat formations are fairly consistent, including a reliable set of five (Table 3, kku) measured in drill core from the top 110 m of the Khok Kruat Formation (middle Cretaceous) at Nakhon Phanom (Fig. 1). Paleolatitudes of individual formations of the Khorat Group (Table 3) differ from present day latitude by $+7.5$ to -7.9 degrees (paleolatitude minus present day latitude), show no trend, and have a mean difference from present day latitude of 0.6 ± 4 degrees. South China Jurassic and Cretaceous paleolatitude differences (Table 3) are much larger, and T-tests show that their differences from Northeast Thailand values are significant to the 99.9 and 98% levels respectively, while their difference from each other is significant to the 90% level.

It is concluded that Northeast Thailand was at nearly the same latitude as now throughout the Jurassic and Cretaceous, and probably also during the late Triassic, although it might have been slightly farther north then (Table 3). In contrast, South China has moved southward by 20 ± 5 degrees since some time in the Jurassic and part of the movement probably occurred in late Jurassic to early Cretaceous time. The change in latitude of South China cannot be accounted for by a 37 degree clockwise rotation of Indochina and South China as a rigid block. In the light of known geology the most likely explanation is large scale dextral displacement along the Red River Fault Zone (Fig. 5) which lies along the Song Ma Geosuture at the south-western edge of South China and is dextral at present (Nguyen Din' Kat 1972, Molnar and Tapponnier 1975). We conclude that dextral strike-slip displacement may have been occurring along the Song Ma Geosuture continuously since Mesozoic time.

3. *Lateral displacement relative to Eurasia*

The Huai Hin Lat (Norian) pole position (Table 2) determined by Achache and Courtillot combined with younger Khorat Group pole positions determined by other authors (Table 1; Table 2) when compared with Eurasian Mesozoic pole positions (Irving 1977) shows convergence of Indochina and Eurasia in the time between late Triassic and middle Jurassic (Achache and Courtillot 1985). According to Achache and Courtillot initial contact of the continental domains occurred before Norian and resulted in anticlockwise differential rotation of Indochina relative to Eurasia. The main differential movement is shown by paleolatitude differences. While Indochina remained at nearly the same latitude as at present, Eurasia moved southward by 1600 ± 750 km (Achache and Courtillot 1985, Fig. 5).

Paleobiogeographic and paleomagnetic evidence (Achache and Courtillot 1985) and lithostratigraphic evidence (above, previous section) leave little doubt that Indochina was sutured to Shan-Thai and South China by late Triassic time. Achache and Courtillot consider that the convergence of the Indochina–Shan-Thai–South China collage with Eurasia between late Triassic and middle Jurassic was centred on the Songpan Ganzai (Bakan Hai Range) which lies between North China (southern edge of Eurasia Plate) and the South China and Tibet paleoplates. They are uncertain whether the convergence resulted from subduction of a remnant of the Paleothethys Ocean or from shortening of continental crust, by either overthrusting or stike-slip faulting.

The clockwise rotation of Northeast Thailand indicated by all Mesozoic declinations (Table 1; Table 2) happened after middle Cretaceous (Maranate 1982b) and is considered by Achache and Courtillot (1985) to have been caused by the collision of peninsula India against Asia in the Cenozoic Era and to have been associated with the suggested propagating extrusion

tectonics of Tapponnier *et al.* (1982). Above (previous section) we have inferred from the paleomagnetism that major dextral displacement has occurred along the Red River Fault in Mesozoic and Cenozoic times, which is contrary to the sinistral displacement suggested by the propagating extrusion hypothesis (Tapponnier *et al.* 1982).

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